

Matrix Operations

Outline

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- Pivoting

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Elementary Row and Column Operations

Introduction: Elementary Row and Column Operations

- ▶ Elementary row and column operations are fundamental tools in linear algebra.
- ▶ They are used in many algorithms, including Gaussian elimination, computation of the inverse matrix, and matrix factorizations (such as the LU decomposition).
- ▶ In this section we define the three elementary row operations and the three elementary column operations, and we see them on examples.

Elementary row operations

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **elementary row operations** are:

1. **Row swap:** Exchange rows i and k .

$$R_i \leftrightarrow R_k$$

2. **Row scaling:** Multiply row i by a nonzero scalar $\lambda \neq 0$.

$$R_i \leftarrow \lambda R_i$$

3. **Row addition:** Replace row i by the sum of row i and λ times row k (with $i \neq k$).

$$R_i \leftarrow R_i + \lambda R_k$$

Example

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After swapping rows 1 and 3, i.e., $R_1 \leftrightarrow R_3$, we obtain:

$$\begin{pmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}.$$

Example

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After multiplying row 2 by $\lambda = -2$, i.e., $R_2 \leftarrow -2R_2$, we obtain:

$$\begin{pmatrix} 1 & 2 & 3 \\ -8 & -10 & -12 \\ 7 & 8 & 9 \end{pmatrix}.$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After replacing row 2 with row 2 minus 4 times row 1, i.e., $R_2 \leftarrow R_2 - 4R_1$, we obtain:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Example (2/2)

Indeed, computing the new row 2:

$$\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ -6 \end{pmatrix}.$$

Elementary column operations

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **elementary column operations** are:

1. **Column swap:** Exchange columns j and k .

$$C_j \leftrightarrow C_k$$

2. **Column scaling:** Multiply column j by a nonzero scalar $\lambda \neq 0$.

$$C_j \leftarrow \lambda C_j$$

3. **Column addition:** Replace column j by the sum of column j and λ times column k (with $j \neq k$).

$$C_j \leftarrow C_j + \lambda C_k$$

Example

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After swapping columns 1 and 3, i.e., $C_1 \leftrightarrow C_3$, we obtain:

$$\begin{pmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{pmatrix}.$$

Example

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After multiplying column 2 by $\lambda = 3$, i.e., $C_2 \leftarrow 3C_2$, we obtain:

$$\begin{pmatrix} 1 & 6 & 3 \\ 4 & 15 & 6 \\ 7 & 24 & 9 \end{pmatrix}.$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

After replacing column 3 with column 3 plus 2 times column 1, i.e., $C_3 \leftarrow C_3 + 2C_1$, we obtain:

$$\begin{pmatrix} 1 & 2 & 5 \\ 4 & 5 & 14 \\ 7 & 8 & 23 \end{pmatrix}.$$

Example (2/2)

Indeed, computing the new column 3:

$$\begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 5 \\ 14 \\ 23 \end{pmatrix}.$$

Summary

Elementary row operations:

- ▶ Row swap $R_i \leftrightarrow R_k$, row scaling $R_i \leftarrow \lambda R_i$ ($\lambda \neq 0$), row addition $R_i \leftarrow R_i + \lambda R_k$ ($i \neq k$)

Elementary column operations:

- ▶ Column swap $C_j \leftrightarrow C_k$, column scaling $C_j \leftarrow \lambda C_j$ ($\lambda \neq 0$), column addition $C_j \leftarrow C_j + \lambda C_k$ ($j \neq k$)

Effect on the Determinant

Introduction: Effect on the Determinant

- ▶ Elementary operations affect the determinant in predictable ways.
- ▶ We see how each row operation and each column operation changes the determinant of a square matrix.
- ▶ These rules are crucial for computational algorithms: they allow us to compute determinants by Gaussian elimination.

Effect of row operations on determinant

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The elementary **row operations** affect the determinant as follows:

1. **Row swap:** $R_i \leftrightarrow R_k$ changes the sign of the determinant.

$$\det(\text{new matrix}) = -\det(\mathbf{A})$$

2. **Row scaling:** $R_i \leftarrow \lambda R_i$ (with $\lambda \neq 0$) multiplies the determinant by λ .

$$\det(\text{new matrix}) = \lambda \det(\mathbf{A})$$

3. **Row addition:** $R_i \leftarrow R_i + \lambda R_k$ (with $i \neq k$) does not change the determinant.

$$\det(\text{new matrix}) = \det(\mathbf{A})$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Example (2/2)

After swapping rows 1 and 2, i.e., $R_1 \leftrightarrow R_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 3 \cdot 2 - 4 \cdot 1 = 2 = -\det(\mathbf{A}).$$

The sign changed as expected.

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 0 = 6.$$

Example (2/2)

After multiplying row 1 by $\lambda = 2$, i.e., $R_1 \leftarrow 2R_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 4 & 2 \\ 0 & 3 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 4 \cdot 3 - 2 \cdot 0 = 12 = 2 \cdot \det(\mathbf{A}) = \lambda \det(\mathbf{A}).$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Example (2/2)

After replacing row 2 with row 2 minus 3 times row 1, i.e., $R_2 \leftarrow R_2 - 3R_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot (-2) - 2 \cdot 0 = -2 = \det(\mathbf{A}).$$

The determinant is unchanged.

Effect of column operations on determinant

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The elementary **column operations** affect the determinant as follows:

1. **Column swap:** $C_j \leftrightarrow C_k$ changes the sign of the determinant.

$$\det(\text{new matrix}) = -\det(\mathbf{A})$$

2. **Column scaling:** $C_j \leftarrow \lambda C_j$ (with $\lambda \neq 0$) multiplies the determinant by λ .

$$\det(\text{new matrix}) = \lambda \det(\mathbf{A})$$

3. **Column addition:** $C_j \leftarrow C_j + \lambda C_k$ (with $j \neq k$) does not change the determinant.

$$\det(\text{new matrix}) = \det(\mathbf{A})$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Example (2/2)

After swapping columns 1 and 2, i.e., $C_1 \leftrightarrow C_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 2 \cdot 3 - 1 \cdot 4 = 2 = -\det(\mathbf{A}).$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Example (2/2)

After multiplying column 2 by $\lambda = 3$, i.e., $C_2 \leftarrow 3C_2$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 6 \\ 3 & 12 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot 12 - 6 \cdot 3 = -6 = 3 \cdot (-2) = \lambda \det(\mathbf{A}).$$

Example (1/2)

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The determinant is:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Example (2/2)

After replacing column 2 with column 2 plus 2 times column 1, i.e., $C_2 \leftarrow C_2 + 2C_1$, we obtain:

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}.$$

The determinant of the new matrix is:

$$\det(\mathbf{A}') = 1 \cdot 10 - 4 \cdot 3 = -2 = \det(\mathbf{A}).$$

Important observation:

- ▶ These properties follow from the fact that the determinant of a matrix is equal to the determinant of its transpose (see the properties of determinants in the chapter on matrices), and a column operation on a matrix is the corresponding row operation on its transpose. Hence column operations have the same effect on the determinant as the corresponding row operations.
- ▶ These results are fundamental for understanding how determinants behave under elementary operations, which is crucial in algorithms such as Gaussian elimination and matrix factorization.

Summary

Effect on the determinant (rows and columns):

- ▶ **Swap:** changes the sign of the determinant
- ▶ **Scaling** by $\lambda \neq 0$: multiplies the determinant by λ
- ▶ **Addition** of a multiple of another row/column: does not change the determinant

Column operations behave as row operations, since $\det$ of a matrix equals $\det$ of its transpose.

Gaussian Elimination and Row Echelon Form



Introduction: Gaussian Elimination and Row Echelon Form

- ▶ Elementary row operations can be used to transform any matrix into a matrix with a “staircase” structure.
- ▶ From this structure many properties of the original matrix (for example, its determinant) can be read easily.
- ▶ The systematic procedure that does this is called **Gaussian elimination**.

Definition 1: Row echelon form

A matrix $\mathbf{U} \in \mathbb{R}^{m \times n}$ is in **row echelon form** if:

1. all its zero rows (if any) are below all its nonzero rows;
2. the first nonzero entry of each nonzero row (called the **pivot** of the row) is strictly to the right of the pivot of the row above it.

- ▶ As a consequence, all the entries below a pivot are equal to zero.
- ▶ A square matrix in row echelon form is upper triangular.

Gaussian Elimination (1/2)

Given $\mathbf{A} \in \mathbb{R}^{m \times n}$, we process the columns from left to right. At each step, considering the rows not yet used as pivot rows:

1. find the first column containing a nonzero entry in these rows; if needed, bring such an entry to the top of these rows with a row swap $R_k \leftrightarrow R_i$ (this nonzero entry is the pivot);

Gaussian Elimination (2/2)

2. for each row i below the pivot row k , create a zero below the pivot with the row addition

$$R_i \leftarrow R_i - \frac{a_{ij}}{a_{kj}} R_k,$$

where a_{kj} is the pivot and a_{ij} is the entry to be eliminated (in the current matrix);

3. repeat on the remaining rows.

At the end, the matrix is in row echelon form. Only row swaps and row additions are used.

Observation 1: Determinant by Gaussian elimination

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\mathbf{U}$ be a row echelon form of $\mathbf{A}$ obtained by Gaussian elimination with s row swaps (and any number of row additions). Then

$$\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii}.$$

In particular, $\det(\mathbf{A}) = 0$ if and only if $\mathbf{U}$ has a zero on its diagonal.

Proof

- Each row addition does not change the determinant and each row swap changes its sign. Hence

$$\det(\mathbf{U}) = (-1)^s \det(\mathbf{A}),$$

$$\text{i.e., } \det(\mathbf{A}) = (-1)^s \det(\mathbf{U}).$$

- Since $\mathbf{U}$ is square and in row echelon form, it is upper triangular, and its determinant is the product of its diagonal entries:

$$\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii}.$$



Example (1/3)

Let us consider

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}.$$

We reduce $\mathbf{A}$ to row echelon form with Gaussian elimination.

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

Example (2/3)

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_2)$$

Example (3/3)

The matrix is now in row echelon form. We used $s = 1$ row swap, so

$$\det(\mathbf{A}) = (-1)^1 \cdot 1 \cdot 1 \cdot 3 = -3.$$

Indeed, by Laplace expansion along the first row:

$$\det(\mathbf{A}) = 0 \cdot (3 - 1) - 1 \cdot (3 - 2) + 2 \cdot (1 - 2) = -3.$$

Summary

Row echelon form:

- ▶ Zero rows at the bottom; each pivot strictly to the right of the pivot above
- ▶ A square matrix in row echelon form is upper triangular

Gaussian elimination:

- ▶ Columns from left to right: choose a pivot (row swap if needed), then $R_i \leftarrow R_i - \frac{a_{ij}}{a_{kj}} R_k$ below it
- ▶ Only row swaps and row additions are used
- ▶ $\det(\mathbf{A}) = (-1)^s \prod_i u_{ii}$, with s the number of row swaps

Pivoting and Partial Pivoting

Introduction: Pivoting and Partial Pivoting

- ▶ In numerical algorithms such as matrix factorizations and row reduction methods, pivoting is a technique used to improve numerical stability and avoid division by zero.
- ▶ The pivot element is the element used as the divisor in the elimination process.
- ▶ We first see when a row swap is **necessary** (zero pivot), and then a strategy to choose the **best** pivot (partial pivoting).

Pivoting

Pivoting is the process of selecting a suitable pivot element in a matrix to perform elimination steps in row reduction algorithms.

When performing row elimination on a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, at step k :

- ▶ The **pivot element** is the entry a_{kk} (the diagonal element in position (k, k)).
- ▶ If $a_{kk} = 0$, we cannot use it as a divisor, so we must swap row k with a row $i > k$ such that $a_{ik} \neq 0$.
- ▶ This row swap is called **pivoting**.

Example (1/2)

Consider the matrix:

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -1 & 0 \\ 2 & 1 & 3 \end{pmatrix}.$$

We cannot use $a_{11} = 0$ as the first pivot. We need to swap row 1 with a row that has a nonzero element in the first column.

We swap rows 1 and 2 (i.e., $R_1 \leftrightarrow R_2$):

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}.$$

Example (2/2)

Now $a_{11} = 1 \neq 0$ and we can proceed with the elimination:

$$R_3 \leftarrow R_3 - 2R_1 : \quad \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & 3 & 3 \end{pmatrix}.$$

Now we use $a_{22} = 2$ as the second pivot:

$$R_3 \leftarrow R_3 - \frac{3}{2}R_2 : \quad \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}.$$

The matrix is now in upper triangular form.

Partial Pivoting

Partial pivoting is a strategy to improve numerical stability by choosing the largest available pivot in absolute value.

At step k of Gaussian elimination:

1. Find the row $i \geq k$ such that $|a_{ik}|$ is maximum among all $|a_{jk}|$ for $j \geq k$.
2. Swap row k with row i (i.e., $R_k \leftrightarrow R_i$).
3. Use the new a_{kk} as the pivot.

Partial Pivoting (continued)

This ensures that:

- ▶ The pivot is the largest element in absolute value in its column, among the entries on or below the diagonal.
- ▶ Division by a larger number reduces round-off errors in floating-point arithmetic.
- ▶ The method is numerically more stable.

Example (1/4)

Consider the matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & -1 & 2 \\ 2 & 3 & -1 \end{pmatrix}.$$

Step 1: Find the largest element in absolute value in the first column.

$$|a_{11}| = 1, \quad |a_{21}| = 3, \quad |a_{31}| = 2.$$

The maximum is $|a_{21}| = 3$, so we swap rows 1 and 2:

$$R_1 \leftrightarrow R_2 : \begin{pmatrix} 3 & -1 & 2 \\ 1 & 2 & 1 \\ 2 & 3 & -1 \end{pmatrix}.$$

Example (2/4)

Now eliminate below the pivot $a_{11} = 3$:

$$R_2 \leftarrow R_2 - \frac{1}{3}R_1 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{7}{3} & \frac{1}{3} \\ 2 & 3 & -1 \end{pmatrix},$$

$$R_3 \leftarrow R_3 - \frac{2}{3}R_1 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{7}{3} & \frac{1}{3} \\ 0 & \frac{11}{3} & -\frac{7}{3} \end{pmatrix}.$$

Example (3/4)

Step 2: Find the largest element in absolute value in the second column (rows 2 and 3).

$$\left| \frac{7}{3} \right| = \frac{7}{3}, \quad \left| \frac{11}{3} \right| = \frac{11}{3}.$$

The maximum is $\left| \frac{11}{3} \right|$, so we swap rows 2 and 3:

$$R_2 \leftrightarrow R_3 : \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{11}{3} & -\frac{7}{3} \\ 0 & \frac{7}{3} & \frac{1}{3} \end{pmatrix}.$$

Example (4/4)

Eliminate below the pivot $a_{22} = \frac{11}{3}$:

$$R_3 \leftarrow R_3 - \frac{7}{11}R_2 : \quad \begin{pmatrix} 3 & -1 & 2 \\ 0 & \frac{11}{3} & -\frac{7}{3} \\ 0 & 0 & \frac{20}{11} \end{pmatrix}.$$

The matrix is now in upper triangular form (row echelon form).

Example (1/4)

Consider the matrix:

$$\mathbf{A} = \begin{pmatrix} \frac{1}{10000} & 1 \\ 1 & 1 \end{pmatrix}.$$

Without partial pivoting:

Using $a_{11} = \frac{1}{10000}$ as the pivot:

$$R_2 \leftarrow R_2 - 10000 R_1.$$

Example (2/4)

This gives:

$$\begin{pmatrix} \frac{1}{10000} & 1 \\ 0 & 1 - 10000 \end{pmatrix} = \begin{pmatrix} \frac{1}{10000} & 1 \\ 0 & -9999 \end{pmatrix}.$$

Problem: The large multiplier 10000 can cause significant round-off errors in floating-point arithmetic.

Example (3/4)

With partial pivoting:

Since $|a_{21}| = 1 > |a_{11}| = \frac{1}{10000}$, we swap rows:

$$R_1 \leftrightarrow R_2 : \begin{pmatrix} 1 & 1 \\ \frac{1}{10000} & 1 \end{pmatrix}.$$

Example (4/4)

Now eliminate using $a_{11} = 1$ as the pivot:

$$R_2 \leftarrow R_2 - \frac{1}{10000} R_1 : \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 - \frac{1}{10000} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & \frac{9999}{10000} \end{pmatrix}.$$

Advantage: The multiplier $\frac{1}{10000}$ is much smaller, leading to better numerical stability.

Practical importance:

- ▶ Partial pivoting is crucial in numerical linear algebra to ensure that algorithms produce accurate results, especially when dealing with matrices that have a wide range of magnitudes.
- ▶ Most modern numerical software (such as MATLAB, NumPy, CPLEX, Gurobi) automatically uses partial pivoting in Gaussian elimination and LU factorization.
- ▶ The computational cost of partial pivoting is minimal compared to the benefit of improved numerical stability.

Summary

Pivoting:

- ▶ If the pivot $a_{kk} = 0$, swap row k with a row $i > k$ such that $a_{ik} \neq 0$

Partial pivoting:

- ▶ At step k , swap row k with the row $i \geq k$ maximizing $|a_{ik}|$
- ▶ Smaller multipliers, fewer round-off errors, better numerical stability

Key Takeaways

Summary - Part 1: Elementary Operations

1. Elementary Row Operations:

- ▶ Row swap: $R_i \leftrightarrow R_k$
- ▶ Row scaling: $R_i \leftarrow \lambda R_i$ (with $\lambda \neq 0$)
- ▶ Row addition: $R_i \leftarrow R_i + \lambda R_k$

2. Elementary Column Operations:

- ▶ Column swap: $C_j \leftrightarrow C_k$
- ▶ Column scaling: $C_j \leftarrow \lambda C_j$ (with $\lambda \neq 0$)
- ▶ Column addition: $C_j \leftarrow C_j + \lambda C_k$

These are fundamental tools used in Gaussian elimination, matrix inversion, and factorizations.

Summary - Part 2: Effect on Determinant

3. Effect on Determinant:

For both row and column operations:

- ▶ **Swap:** Changes sign of determinant ($\det \rightarrow -\det$)
- ▶ **Scaling:** Multiplies determinant by scalar ($\det \rightarrow \lambda \det$)
- ▶ **Addition:** Does not change determinant ($\det \rightarrow \det$)

These properties are essential for understanding numerical algorithms and matrix factorizations.

Summary - Part 3: Gaussian Elimination

4. Row echelon form and Gaussian elimination:

- ▶ Row echelon form: zero rows at the bottom, pivots moving strictly to the right
- ▶ Gaussian elimination reduces any matrix to row echelon form with row swaps and row additions
- ▶ $\det(\mathbf{A}) = (-1)^s \prod_i u_{ii}$ (s = number of row swaps)

Summary - Part 4: Pivoting Strategies

5. Pivoting:

- ▶ Process of selecting suitable pivot elements in elimination algorithms
- ▶ Necessary when diagonal element is zero
- ▶ Involves row swaps to bring nonzero element to pivot position

6. Partial Pivoting:

- ▶ Choose the largest element (in absolute value) as pivot
- ▶ Improves numerical stability by reducing round-off errors
- ▶ Standard practice in modern numerical software
- ▶ Minimal computational cost with significant benefits